

Fannie Mae Single-Family Loan Performance Data

Personal Reference & Pipeline Notes

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■ **Purpose:** This document is your quick-reference guide to what Fannie Mae's Single-Family Loan Performance dataset actually contains, how it is structured, and where it differs from the proprietary CoreLogic data used in Sirignano et al. (2021). It also flags data-quality quirks that matter for pipeline design. Keep it alongside the `00_OVERVIEW.md` and `01_SCHEMA.md` files.

1. What Is This Dataset?

Fannie Mae's **Single-Family Loan Performance (SFLP)** dataset is a publicly available research dataset that contains origination and monthly performance records for a subset of single-family mortgage loans acquired by Fannie Mae. It was first released in April 2013 and is updated quarterly, typically about four months after the end of the most recent quarter.

The data are organized as a **loan-month panel**: one row per loan per monthly reporting period, from acquisition through termination (payoff, foreclosure, REO disposal) or the data cut-off date. This structure is exactly what you need for a competing-risks survival model — each row is one "spell" in a given state, and the transition to the next state is the supervised target.

Your slice of the data:

Property	Value
Format	Pipe-delimited CSV, no header row
Fields per row	113 (positional — column order is the contract)
Grain	One row per loan × monthly reporting period
Total raw size	~219 GB across 26 quarterly files
Largest file	2020Q4.csv ≈ 25.4 GB
Date encoding	MMYYYY (e.g., 112013 = November 2013)
Coverage (your files)	2005Q4, 2013Q4–2014Q3, 2017Q2–2022Q2 (gaps noted)
Compressed Parquet estimate	~30–50 GB (4–8× compression with zstd)

■■ **Coverage Warning:** Your raw files have large gaps — no 2006–2012, no 2015–2016, no 2017Q1 — and at least three truncated files (2013Q4 has only 15 distinct loans). The pipeline's Stage 1 inventory audit is *mandatory* before any modelling work to prevent these partials from silently corrupting the panel.

2. Population & Eligibility Criteria

The dataset is **not** a census of all Fannie Mae loans — it is a curated research sample. Understanding what is and is not included is critical for correctly interpreting any model results.

What is included (primary dataset):

Criterion	Value
Origination date	On or after January 1, 1999
Acquisition date	On or after January 1, 2000 (updated quarterly)
Product	Fixed-rate, fully amortizing mortgages only
Term	≥ 5 years and < 35 years at origination
Documentation	Full documentation loans only
Maturity	30-year and shorter conventional terms

What is excluded (primary dataset):

Exclusion	Reason / Impact
HARP / Refi Plus loans	These have a dedicated separate dataset
Adjustable-rate mortgages (ARMs)	Rate reset dynamics excluded
Interest-only loans	Non-standard amortization
Balloon mortgages	Non-standard amortization
Prepayment penalty loans	Non-standard prepayment incentives
LTV > 97%	High-risk product removed for data quality
Alt-A / reduced documentation	Data quality and model comparability
Government-insured (FHA, VA)	Fannie Mae conventional only
Long-term standby commitments	Operational exclusion
Lender recourse / risk-sharing*	Except primary MI — counterparty complexity
Bulk/negotiated acquisitions	Non-standard pricing
Odd due dates / bi-weekly	Non-standard payment timing
Pre-1999 originations	Incomplete historical records
Same-month acquisition + liquidation	Operationally degenerate observations

* Loans in Credit Risk Transfer (CRT) programs — CAS, CIRT, EPMI, Seller/Servicer Risk Share — are *not* excluded despite risk-sharing, per FAQ §3 footnote 4.

■ **Modelling implication:** Because ARMs, IO loans, and prepayment-penalty loans are excluded, your Fannie Mae model will capture a *cleaner, more homogeneous* product than Sirignano et al.'s CoreLogic data (which includes all product types). This means your prepayment nonlinearities may look different — most of the exotic prepayment-incentive effects in the paper come from ARMs and hybrid products.

3. Key Variables and What They Mean

The 113-column file layout is documented in *01_SCHEMA.md*. Below is a functional grouping of the fields that matter most for the Sirignano-style competing-risks model.

Group A: Origination / Static Features (don't change over the loan's life)

Field	Type	Notes
Original Interest Rate	f32	Rate at closing. Core input for prepayment incentive (rate – PMMS).
Original UPB	f32	Unpaid principal balance at origination. Log-transform before use.
Original Loan Term	i16	Months (360 = 30yr, 180 = 15yr, etc.).
Original LTV	f32	LTV at origination. Note: CLTV separately captures subordinate liens.
Original CLTV	f32	Combined LTV including all liens. More complete credit-risk signal.
DTI	f32	Debt-to-income at origination. Blank in HARP (see §5).
FICO (Borrower)	i16	Sentinel 9999 → null. Separate co-borrower FICO from Jan 2015+.
FICO (Co-Borrower)	i16	Lower of the two was disclosed pre-2015; both since Jan 2015.
Origination Date	date	MMYYYY. Defines vintage. Encodes as YYYYMM integer for filters.
First Payment Date	date	One month after origination; use for loan-age calculation.
First-Time Home Buyer	cat	Y/N indicator. Affects default behavior.
Loan Purpose	cat	P=Purchase, C=Cash-Out Refi, R=Rate-Term Refi, U=Unknown.
Property Type	cat	SF/CO/CP/MH/PU. Single-family dominates.
Occupancy Status	cat	P=Primary, S=Second Home, I=Investment/Non-Owner.
Property State	cat	2-letter. Proxy for legal foreclosure regime.
MSA	cat	5-digit code. '00000' = not in any MSA. Join to HPI series.
Zip Code Short	cat	First 3 digits only (privacy truncation). Join to PMMS rates.
Channel	cat	R=Retail, B=Broker, C=Correspondent. Related to prepayment behavior.
Amortization Type	cat	FRM/ARM. In primary dataset this should be FRM-only.
Interest-Only Indicator	cat	Y/N. Should be all-N in primary dataset (IO excluded).
MI Percentage	f32	Mortgage insurance coverage %. Affects credit risk exposure.

Group B: Dynamic / Performance Features (update each month)

Field	Type	Notes
Monthly Reporting Period	date	MMYYYY. The time index of the panel.
Current Interest Rate	f32	Changes post-modification. Compute rate incentive = orig – current PMMS.
Current Actual UPB	f32	Loan balance this month. Post-Oct 2020: reports 0 at liquidation.
Loan Age	i16	Months since first payment. Can be –1 if acquired before first payment.
Remaining Months to Legal Maturity	i16	Contractual remaining term; persists through modification.
Remaining Months to Maturity	i16	Blank for modified loans (use Legal Maturity instead).
Current Loan Delinquency Status	str	00=Current, 01=30DPD, 02=60DPD, 03+=90+DPD, XX=Unknown. PRIMARY TARGET DRIVER.
Loan Payment History	str	24-month string of delinquency codes. Burnout / recidivism features.
Modification Flag	cat	Y/N. Can only track first modification; does not increment.
Servicing Activity Indicator	cat	Flags servicer transfers that may artifactually inflate delinquency. Filter carefully.
Unscheduled Principal Current	f32	Curtailment signal. High values → partial prepayment.
Zero Balance Code	cat	Present ONLY in termination month. See §4. SECONDARY TARGET DRIVER.

Group C: Termination Fields (appear at/after loan exit)

Field	Type	Notes
Zero Balance Code	cat	01=Prepay/Matured, 02=Third-party sale, 03=Short sale, 06=Repurchase, 09=REO disposition, 15=Note sale, 16=RPL sale, 96=Non-credit removal, 97/98=Credit repurchase.
Zero Balance Effective Date	date	Date balance went to zero. Liquidation date for codes 01, 06, 16.
Foreclosure Date	date	Month+year only (day masked for privacy). Date loan entered foreclosure process.
Disposition Date	date	Month+year only. Date Fannie Mae's interest in property ended (REO sale, short sale, etc.).
UPB at Time of Removal	f32	Starting Oct 2020, captures UPB at liquidation (replaces UPB=0 convention).
Net Sales Proceeds	f32	Proceeds from property sale. Null = 0 for this field per FAQ §46.
Foreclosure Costs	f32	Typically populated ~90 days after Disposition Date.
Credit Enhancement Proceeds	f32	MI claim payments + other credit enhancement proceeds.

4. The Seven-State Target (Sirignano Mapping)

The model predicts the one-month-ahead **state transition**. Each loan-month is assigned to one of seven states derived from two fields: **Current Loan Delinquency Status** (col 39) and **Zero Balance Code** (col 43). The Zero Balance Code only appears in the loan's final (terminating) month; in all prior months it is blank.

State	Source Field	Condition	Meaning
current	Delinq. Status	== '00'	Loan is current
dpd_30	Delinq. Status	== '01'	30 days past due (1 missed payment)
dpd_60	Delinq. Status	== '02'	60 days past due
dpd_90plus	Delinq. Status	>= '03'	90+ days past due (also 120+)
foreclosure	Zero Bal. Code	02/03/15/97/98	Credit-loss termination events
REO	Zero Bal. Code	09	Real estate owned — Fannie Mae holds property
prepaid	Zero Bal. Code	01/06/16/96	Full payoff, repurchase, or non-credit removal

Derivation priority: if Zero Balance Code is present → use it; otherwise use delinquency status. Delinquency code 'XX' (unavailable) → null. The **state_next** column is derived by leading the state by one reporting period within each loan, using a window function ordered by reporting month.

■ **Leakage rule for rolling backtests:** Because *state_next* looks one month ahead, a training window ending at calendar month T must mask on `period_ym < T` (so the labelled next-month is $\leq T$). Using `period_ym ≤ T` would leak the next month's event into training. The integer `period_ym` column (YYYYMM) makes this a one-line DuckDB filter.

5. Primary Dataset vs. HARP Dataset

Fannie Mae publishes two distinct datasets under the SFLP umbrella. You are working with the **primary dataset only**. Understanding what the HARP dataset is and why it exists helps clarify why HARP loans are absent from your data.

Dimension	Primary Dataset	HARP Dataset
Scope	Core research population: fixed-rate fully-amortizing conventional loans acquired from Jan 2000.	Subset of ~1 million primary-dataset loans that were subsequently refinanced through HARP (Apr 2009 – Sep 2016).
Population size	Several tens of millions of loans across all cohorts.	~1 million loans (fixed subset; no new HARP acquisitions expected).
Loan products	30-year and shorter fixed-rate, fully amortizing only.	30-year, 20-year, and 15-year fixed-rate (HARP refinanced loan terms).

Dimension	Primary Dataset	HARP Dataset
Origination window	Jan 1, 1999 onwards (acquired from Jan 2000).	Original loan closed on or before May 31, 2009; acquired by Fannie Mae Jan 2000 – Sep 2015.
DTI field	Populated where available.	BLANK for all loans — HARP did not require full re-underwriting; no updated income data.
LTV calculation	Standard: original balance ÷ appraised value at origination.	AVM-based if Fannie Mae waived fieldwork requirements; appraisal-based otherwise.
Loan ID continuity	Single stable loan ID throughout.	Loan ID changes at the moment of HARP refinance; a mapping key file links old ID to new ID.
What happens at HARP refinance (in primary data)	HARP refinance appears as a prepayment (Zero Balance Code = 01). Loan exits primary dataset at that point.	HARP dataset picks up where the primary dataset left off: post-refinance performance is appended to pre-refinance history.
Update cadence	New loan acquisitions added quarterly (4-month lag).	Population is static (no new HARP acquisitions); performance data updated quarterly alongside primary.
Primary purpose	General mortgage credit & prepayment research.	Analysis of HARP program performance; benchmarking for Fannie Mae's high-LTV refinance option (successor to HARP).
Refi Plus relationship	Excludes all Refi Plus loans (including HARP).	HARP loans only (LTV > 80% at origination). Excludes Refi Plus loans with LTV ≤ 80%.

■ **Key takeaway for your project:** In the primary dataset, every HARP refinance looks like a standard prepayment (ZBC = '01'). This means your model will treat those exits as prepayments — which is economically reasonable (the borrower refinanced to a better rate) even though the property and borrower risk did not disappear. If you later want to isolate 'true' prepayments from HARP-driven exits, you would need to join the mapping key from the HARP dataset against your panel. For your dissertation scope this is not necessary.

6. Data Quality Issues and Pipeline Quirks

Delinquency jumps > 1 month

The FAQ explicitly acknowledges that delinquency status can jump by more than one bucket in a single month (e.g. current → 90+ DPD). This is real data, not an error: it can result from misapplied payments, servicer reporting delays, or court actions. Your model must handle this. The seven-state Markov formulation implicitly allows non-adjacent transitions.

Servicer Activity Indicator

Loans flagged here may show a spurious delinquency due to a reporting change inside a servicer, not an actual missed payment. Filter these observations with care in exploratory analysis; for modelling, leaving them in (they are real servicer events) is generally the correct approach, but label them.

Negative loan age (–1)

Loan Age is current period minus first payment date. If Fannie Mae acquired the loan before its first payment was due, this field shows –1. Safe to treat as 0 in features.

Remaining Months to Maturity = blank for modified loans

Post-modification, this field is not populated. Use Remaining Months to Legal Maturity instead — it does reflect the modified contractual term.

UPB = 0 at liquidation (post-Oct 2020)

Starting Oct 2020, Current Actual UPB is reported as 0 when the loan terminates. Pre-Oct 2020 it held the actual balance. Use the new UPB at Time of Removal field for liquidated balances in post-2020 data.

Expenses / proceeds on 90-day lag

Foreclosure costs and sale proceeds are generally populated ~90 days after Disposition Date. For very recent liquidations, these fields will be partial or blank.

Credit scores: single vs. dual (pre-/post-Jan 2015)

Before the January 2015 release, Fannie Mae disclosed only the lower of borrower / co-borrower FICO. From Jan 2015 both are provided. Your schema has both fields; the missingness indicator `fico_co_missing` captures loans with no co-borrower.

Seller/Servicer names aggregated to 'Other'

Any entity representing < 1% of volume in a given quarter is shown as 'Other'. Do not use these fields as high-cardinality ID features without acknowledging this censoring. Safe to use as low-cardinality categorical proxies.

Truncated files in your download

2013Q4 (302 KB, 15 loans), 2014Q3 (~3.9 MB), 2005Q4 (~1.1 MB) are orders of magnitude too small to be real quarters. Stage 1 must catch these and quarantine them — do not mix them into the panel without understanding what they contain.

MSA values updated over time

MSA codes are periodically revised by the US OMB. Fannie Mae updates MSA in the dataset going forward. There may be a lag. If you join HPI data by MSA, be alert to time-varying MSA boundary changes.

No full zip code

Privacy: only 3-digit truncated zip. For PMMS rate incentive, you will need to aggregate or interpolate the FHFA/Freddie Mac series to the 3-digit level.

MBS vs. cash loans indistinguishable

There is no field to tell you whether a loan was securitized or held in portfolio. This does not affect your model but matters if you try to price MBS cash flows.

7. Fannie Mae vs. CoreLogic (Sirignano et al.)

Sirignano, Sadhwani & Giesecke (2021) use a **proprietary CoreLogic dataset** licensed from a third-party data aggregator that collects records directly from originators and servicers. That dataset is not publicly available. Your Fannie Mae data is the closest publicly accessible analog.

Dimension	CoreLogic (Paper)	Fannie Mae SFLP (Your Data)
Loans	~120 million (prime + subprime)	~10s of millions (conventional fixed-rate only)
Loan-month obs.	~3.5 billion	~1.5–2 billion (your files)
Origination period	1995–2014	1999–present (your files: 2005–2022 partial)
Mortgage types	ALL: fixed, ARM, hybrid, balloon, IO, subprime	Fixed-rate, fully amortizing only (no ARM/IO/balloon)
Prime/subprime	Both (25M subprime + 93M prime)	Conventional only (no FHA/VA; effectively prime)
Geographic granularity	Zip-code level HPI (Zillow), county unemployment	3-digit zip, MSA; need external HPI/rate joins
Macro covariates	Built into dataset (Zillow, BLS, Freddie Mac)	Not in dataset; must join FHFA HPI and PMMS externally
DTI field	Available for most loans	Available in primary; blank for all HARP loans
Availability	Proprietary (licensed)	Public (registration required)
Result replicability	Exact replication impossible	Methodology replicable; results will differ

■ **Bottom line:** Your Fannie Mae data covers a *cleaner, narrower* product universe (fixed-rate conventional loans) versus the paper's all-product CoreLogic data. The modeling methodology — deep learning for competing-risks state transitions — transfers completely. The specific nonlinear effects (especially ARM-driven prepayment and subprime default patterns) will look different. This is expected and should be discussed as a data limitation in your dissertation.

8. External Data Required for Feature Engineering

Source	Description	Use in Model	Access
Freddie Mac PMMS	Weekly Primary Mortgage Market Survey rate series	Compute rate incentive = Current Interest Rate – market rate at current period	freddiemac.com/pmms — free public
FHFA HPI	Federal Housing Finance Agency House Price Index	Compute current LTV = original LTV × (original HPI / current HPI) by MSA/state	fhfa.gov/data — free public

These two external series are essential for replicating the key Sirignano et al. features: the **prepayment incentive** (the primary driver of prepayment nonlinearity) and **current LTV** (the primary driver of default). Both are freely available at national and regional levels.